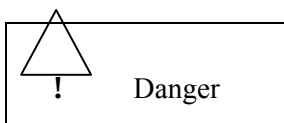


Chapter 1 Safety Rules

1.1 Cautions in Installation



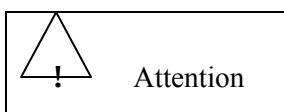
Perform the maintenance job after confirming that the charging LED is off or the DC Bus voltage is below 36V.

Don't touch the live control terminals with bare hands, otherwise electrical shocking may occur.

The earthing terminal of the drive must be earthed reliably, otherwise electrical shocking accident may occur.

Make sure the input AC supply is disconnected totally before doing the wiring job, otherwise electrical shocking may occur.

Only professional personnel can do the wiring job, otherwise electrical shocking may occur.



If over current occurs after restarting the drive, please check if the external wiring is correct before switch on the drive.

Generally, the user should remove the cover of the drive to clean the drive's internal parts periodically with air (generally 1 to 3 months depending on the field condition);

The power cables' bare parts must be bound with insulation tapes, otherwise material loss may be resulted;

It is absolute forbidden to connect the AC 220V supply to control terminals except the terminals TA, TB and TC;

Don't mistake the input terminals (L, N) with the output terminals (U, V, W) or the DC bus terminals with braking terminals P(+), (−) and PB, otherwise the drive may be damaged;

Don't install the drive in the place with direct sunshine, otherwise it may be damaged;

Don't operate the drive whose parts are not complete, otherwise there is a danger of a fire or human injury;

Don't let the screws, washers and other metal foreign matters fall inside the drive, otherwise there is a danger of fire or damaging;

Chapter 2 Motor Drive's Panel and Operate

2.1 Operation Panel and Status Display Units: Fig. 2-1

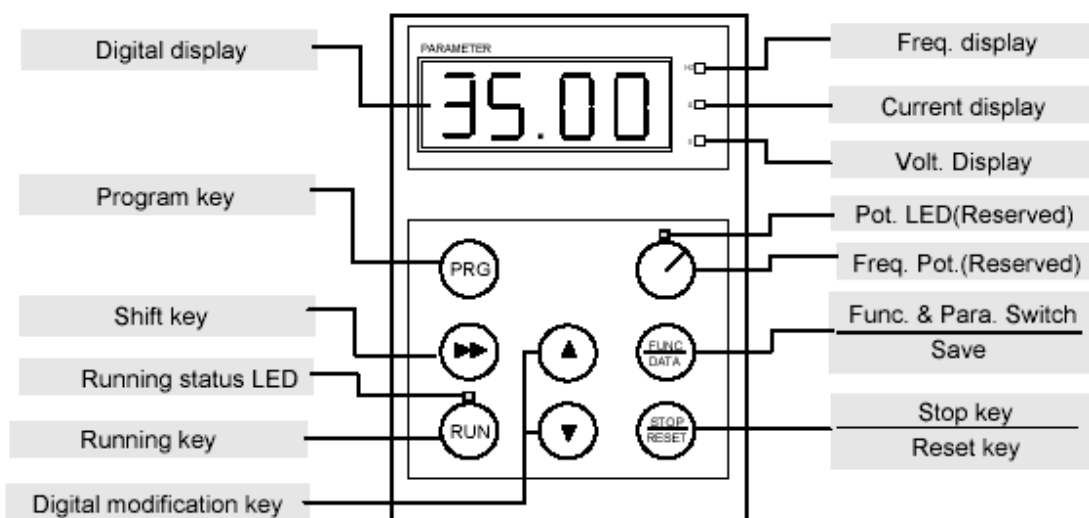


Fig. 2-1 (Operation panel)

Operation Panel Key Functions

Key	Name	Function
PRG	Programming	Switches between stopping status/operating status and programming state.
FUNC/DTAT	Function /Data	Select data monitor mode and data input confirmation
▲	Up	Increase
▼	Down	Decrease
▶▶	Shift	In the status of RUN and STOP, press this key to select the display of parameters; when setting data, press this key to select the setting data's modification bit, or to switch function code by sections
RUN	Run	In the panel control operation mode, press this key to start.
STOP/RESET	Stop/Reset	In the panel control operation mode, press this key to stop operating, or press it to reset and end fault-alarming status.
/	Frequency potentiometer	Reserved function

In the elevator-door controlling manual tuning mode, press **RUN** key and **▲** key together to OD, and press **RUN** key and **▼** key together to CD.

Under the universal drive panel operation mode, press **RUN** key and **▲** key together to run FWD, press **RUN** key and **▼** key together to run REV.

2.2 Func. Code Parameters' Setting Method

Func. Code parameter settings can be done only via the operation panel.

The panel parameter setting method will be introduced in the following example of setting low-speed operating speed from 3Hz to 4Hz(change 3 into 4 in F005).

1. Press **PRG** key - to enter programming state

The LED on the panel will display current function code (eg. F000)

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2. Press **▲**, **▼** key or **SHIFT** key - to shift to the parameter function code needs to be changed (F005)

Note:

If there is no continuous display of function parameters, then should unpack the corresponding function group before any operation

3. Press **FUNC/DATA** key - to shift to the corresponding parameter (3).

4. Press **▶▶** key to shift to the right flashing bit that needs to be changed (3 flashing)

5. Press **▲** or **▼** key - to tune the parameter to the wanted value (Press **▲** key and tune to 4)

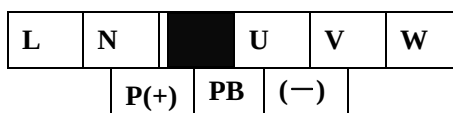
6. Press **FUNC/DATA** key - to save and auto-display next Func.Code (display F006)

7. Press **PRG** key - to exit programming state

Chapter 3 Motor Drive's Wiring and Tuning

3.1 Wiring

1. Power Terminals of Main Circuit



Terminal Name	Function
P(+) 、 PB 、 (—)	P(+): “+” bus; PB: terminal for connecting braking unit, (—): “-” bus;
L 、 N	Input terminals for single-phase 220Vac
U 、 V 、 W	Output terminals for motor
PE	Terminal for earthing

Safety earthing cable should be connected to the screw marked with “”.

2. Control Terminals

PA	PB	PC	PAC	PC1	PC2	P24	COM	X1	X2	X3	X4	X5	X6	X7	COM	OD	CD
-----------	-----------	-----------	------------	------------	------------	------------	------------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	------------	-----------	-----------

Terminal	Terminal Code	Terminal function description	Specification
Output terminals	PA、PB、PC	Programmable relay output 0	Ratings of contacts: AC: 250V/2A; DC: 30V/1A
	PAC、PC1	Programmable relay output 1	
	PAC、PC2	Programmable relay output 2	
	P24	24V power supply (reference GND is COM)	24V power supply (reference GND is COM)
Terminals for inputting OD and CD commands	OD	OD command (reference GND is COM)	24V input, X1 and X2 can meet the input requirements of pulse signal with frequency below 40kHz
	CD	CD command (reference GND is COM)	
Multifunction Input terminal	X1	PG A-phase signal input	
	X2	PG B-phase signal input	
	X3~X7	Functions are programmable (Reference GND is COM)	

3.2 System Wiring Diagram

In distance control, the speed changing and complete CD and OD judgments are made according to the actual pulse count number. System wiring diagram for distance control is shown in Fig. 3-1:

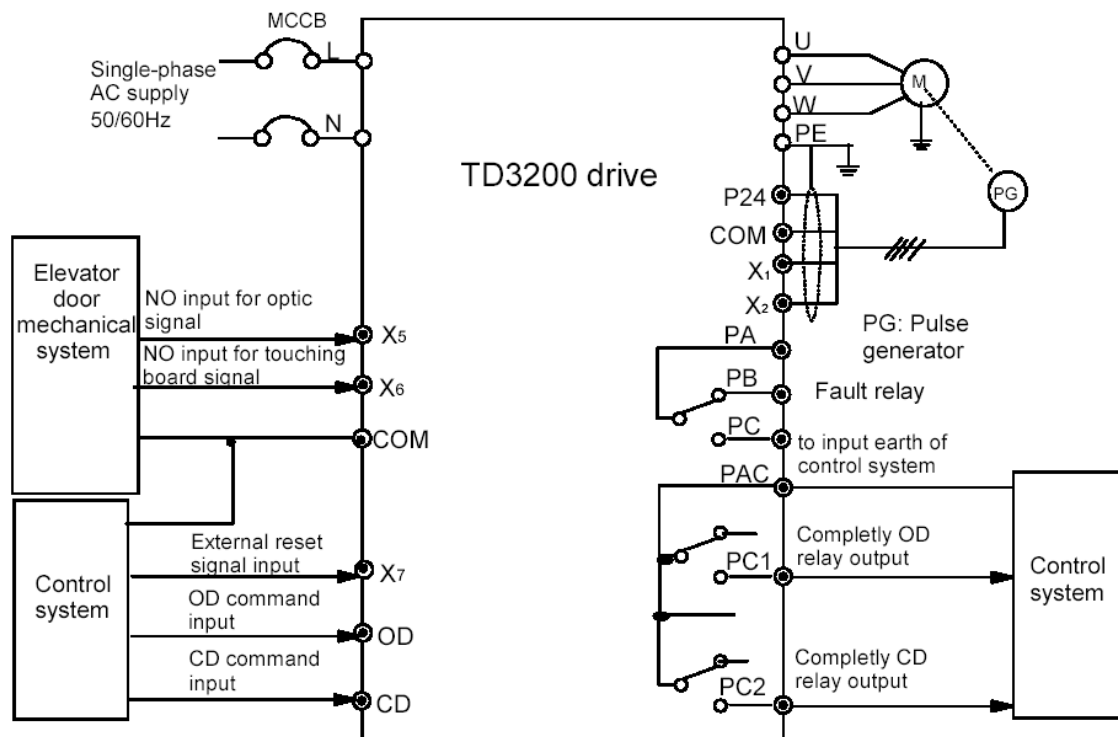


Fig. 3-1 (System wiring diagram for distance control)

PG wiring diagram for distance control is shown in Table. 3-2:

Table. 3-2 PG wiring diagram

Color	Function	Terminal
Brown	+24V	P24
Black	GND	COM
Blue	PG A-phase signal input	X1
Yellow	PG B-phase signal input	X2

3.3 Relevant Functional Parameters

Parameter settings are shown in Table 3-3:

Table. 3-3 Functional parameter settings for Distance control (For example PL=1500 in Eshine system)

Functions	Name	Setting	Remark
F001	Control mode	3	“Distance control 2”
F097	Input 10 Function for digital input terminal X1	10	Phase A PG signal
F098	Input 11 Function for digital input terminal X2	11	Phase B PG signal
F101	Function for digital input terminal X5	2	Normally open input contacts for optic signal
F102	Function for digital input terminal X6	4	Normally open input contacts for touching board

F103	Function for digital input terminal X7	1	External reset (RESET) signal input
F105	Programmable relay PAC/PC1	0	Output signal for complete OD S1
F106	Programmable relay PAC/PC2	1	Output signal for complete CD1 S1
F010	Start torque for OD	25.0%	Parameters in the table are factory settings, adjust the OD parameters according to the actual OD operating conditions. (Show in Fig. 5-3)
F011	OD startup Acc time	0.5s	
F012	Low speed in OD startup process	7.0Hz	
F013	Low-speed maintaining time at OD startup	1.1s	
F014	OD reference frequency setting	42Hz	
F015	OD Acc time	0.8s	
F018	OD Dec time	1.0s	
F021	Low speed setting in OD ending phase	3Hz	
F022	Holding torque in complete OD status	25.0%	
F023	Threshold setting for torque maintaining	22.0%	
F024	Abnormal Dec time	0.5s	Parameters in the table are factory settings, adjust the CD parameters according to the actual CD operating conditions. (Show in Fig. 5-4)
F027	CD start torque	15.0%	
F028	CD startup Acc time	0.5s	
F029	Low speed in CD startup process	4Hz	
F030	Low-speed maintaining time at CD startup	1.0s	
F031	CD reference frequency setting	32Hz	
F032	CD Acc time	0.8s	
F035	CD Dec time	0.8s	
F038	Low speed setting in CD ending phase	4Hz	
F039	Holding torque for complete CD	35.0%	
F040	Hindering torque for CD	50.0%	Set according to actual operating (Show in Fig. 5-2, 5-3, 5-4)
F041	Threshold setting for CD torque changing	50.0%	
F042	Hindering torque setting for CD low speed end phase	70.0%	
F049	OD speed changing location under distance control	76.0%	Set according to actual operating (Show in Fig. 5-2, 5-3, 5-4)
F050	CD speed changing location under distance control	78.0%	
F051	Pulse setting for complete OD under distance control	93.0%	Changeless value
F052	Pulse setting for complete CD under distance control	94.4%	Show in Table3-4、3-5

Table3-4 The setting of F52 for center opening 、 four panels center opening 、 six panels center opening door

Opening sizes	600	700	800	900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000
Setting of F52	87.4%	89.0%	90.2%	91.1%	91.9%	92.6%	93.2%	93.6%	94.1%	94.4%	94.8%	95.0%	95.3%	95.5%	95.8%
Opening sizes	2100	2200	2300	2400	2500	2600	2700	2800	2900	3000					
Setting of F52	95.9%	96.1%	96.3%	96.4%	96.6%	96.7%	96.8%	96.9%	97.0%	97.1%					

Table3-5 The setting of F52 for side opening 、 three panels side opening door

Opening sizes	600	700	800	900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000
Setting of F52	93.2%	94.1%	94.8%	95.3%	95.8%	96.1%	96.4%	96.7%	96.9%	97.1%	97.3%	97.4%	97.6%	97.7%	97.8%

Chapter 4 Function Parameters Table

The parameter of “Factory Setting” is the most reasonable setting before it leaves factory. The parameter of “Initial setting” is the initial value by the driver, User can not modify it .

In the “ Mod.” column of the function tables,

“ O” means that the parameter can be modified during Drive’ s operating status;

“ ×” means that the parameter can not be modified during Drive’ s operating status;

“ —” means that the parameter is set by the door factory and is not recommended to be modified by the user.

4.1 Basic Operation Function Parameters

Func. Code	Name	Setting Range	Factory Setting	Initial Value	Mod.
F000	User password	0~9999 (0000 means no password)	1234	0	○
F001	Control mode	0: Speed control 1 1: Speed control 2 2: Distance control 1 3: Distance control 2	3	0	—
F002	Control commands’ selection	0: Panel control mode of universal drive 1: Terminal control mode of the drive used for controlling elevator door 2: Manual testing mode 3: Auto-demo mode 4: Terminal control mode of universal drive	1	0	X
F003	Max.output frequency	50.00Hz~400.0Hz	50.00Hz	50.00Hz	X
F004	Operating direction selection	0: Conformed to the pre-set direction 1: Reversed to the pre-set direction	0	0	—
F005	Low-speed operation setting	0.00~3600s	7.00Hz	3.00Hz	X
F006	Limit of CD or OD time	0.1~3600s	300.0s	300.0s	—
F007	Delay time for OD terminal command	0.0~3600.0s	3600.0s	0.0s	X
F008	Delay time for CD terminal command	0.0~3600.0s	3600.0s	0.0s	X

4.2 OD/CD Operating Parameters

Func. Code	Name	Setting Range	Factory setting	Initial Value	Mod.
F009	Display selection for F009~F025	0: Not display 1: Display	1	0	○
F010	OD start torque	0.0%~150%(motor rated torque)	25.0%	50.0%	X
F011	OD startup Acc time	0.1 ~3600s	0.5s	1.0s	○
F012	Low speed setting in OD startup process	0.00~50.00Hz	4.0Hz	10.0Hz	X
F013	Low speed maintaining time at OD start	0.1 ~3600s	1.8s	1.0s	○
F014	OD reference frequency setting	0~Max. frequency(Hz)	35-45.00H	35.00Hz	○
F015	OD Acc time	0.1 ~3600s	0.8s	2.0s	○
F018	OD Dec time	0.1 ~3600s	1.0s	2.0s	○
F021	Low speed setting in OD ending p	0.00~50.00Hz	3.00Hz	3.00Hz	X
F022	Holding torque in complete OD st	0.0%~150%(Motor rated torque)	25.0%	50.0%	X
F023	Threshold setting for OD t changing	0.0%~150%(Motor rated torque)	22.0%	50.0%	X
F024	Abnormal Dec time	0.1~100s	0.5s	0.5s	○
F025	Reserved				
F026	Display selection for F026~F042	0: No display of this function module 1: Display of this function module	1	0	○
F027	Start torque for CD	0.0%~150%(Motor rated torque)	15.0%	50.0%	X
F028	CD startup Acc time	0.1 ~3600s	0.5s	1.0s	○
F029	Low speed setting in CD s process	0.00~50.00Hz	4.00Hz	8.00Hz	X
F030	Low speed maintaining time at CD startup	0.1 ~3600s	1.0s	1.0s	○
F031	CD reference frequency setting	0.00~Max. frequency(Hz)	30-40.00H	30.00Hz	○
F032	CD Acc time	0.1 ~3600s	0.8s	2.0s	○
F035	CD Dec time	0.1 ~3600s	0.8s	2.0s	○
F038	Low speed setting in CD ending phase	0.00~50.00Hz	4.00Hz	2.00Hz	X
F039	Holding torque in complete CD status	0.0%~150%(Motor rated torque)	35.0%	50.0%	X
F040	Hindering torque setting for CD speed phase	0.0%~150%(Motor rated torque)	50.0%	100.0%	X
F041	Threshold setting for CD torque changing	0.0%~150%(Motor rated torque)	40.0%	50.0%	X
F042	Hindering torque setting for CD low speed end phase	0.0%~150%(Motor rated torque)	70.0%	100.0%	X

4.3 Distance Control Parameters

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Func. Code	Name	Setting Range	Factory setting	Initial Value	Mod.
F043	Display selection for F043~F066	0: Not display 1: Display	1	1	○
F044	Number of pulse per revolution	1~9999	100	128	—
F045	Number of phases of PG	0: double phase 1: single phase	0	0	—
F046	Direction of PG	0: FWD 1: REV	1	0	—
F047	Speed in door-width auto-learning	0~50.00Hz	7.00Hz	5.00Hz	X
F048	Function selection for door-auto-learning	0, 1. From 0 → 1 to start door width auto-learning, then auto turn to 0 after finishing. Auto learning is effective in motor manual tuning mode	0	0	X
F049	OD speed changing location setting under Distance control	60.0%~90.0%(door-width)	68-86.0%	70.0%	X
F050	CD speed Changing location setting under Distance control	60.0%~90.0%(door-width)	70-78.0%	70.0%	X
F051	Pulse setting for completely OD under Distance control	80.0%~99.0%(door-width)	93.0%	95.0%	X
F052	Pulse setting for completely CD under Distance control	80.0%~99.0%(door-width)	Show in Table 3-4, 3-5	95.0%	X

4.4 Multi-section Speed Parameters

F055~F062 User need not modify it.

4.5 Demo Function Parameters

Func. Code	Name	Setting Range	Factory set	Initial Value	Mod.
F063	Holding time for completely OD in demo mode	1~3600s	5.0s	2.0s	○
F064	Holding time for completely CD in demo mode	1~3600s	5.0s	2.0s	○
F065	CD/OD operation times record in demo mode	0~9999		0	X
F066	Preset CD/OD operation times in demo mode	0~9999		0	X

4.6 Motor' s Parameters

F067~F078 These parameters are the most reasonable setting in door factory.

User need not modify it.

4.7 Auxiliary Parameters

F082~F086 User need not modify it.

4.8 Vector Control Parameters

F087~F095 User need not modify it.

4.9 Digital I/O Function Parameters

Func. Code	Name	Setting Range	Factory setting	Initial Value	Mod.
F096	Display selection F096~ F108	0: Not display 1: Display	1	0	○
F097	Function selection for control terminal X1	0: No function (can be selected again) 1: External reset(RESET)input 2: Normally open input for optic signal	10	0	—
F098	Function selection for control terminal X2	3: Normally closed input for optic signal 4: Normally open input for touching board 5: Normally closed input for touching board 6: OD position control signal normally open input	11		—
F099	Function selection for control terminal X3	7: OD position control signal normally closed input 8: CD position control signal normally open input 9: CD position control signal normally closed input	0		X
F100	Function selection for control terminal X4	10: PG A-phase signal input(X1,X2) 11: PG B-phase signal input(X1,X2) 12: OD speed-shift contact normally open input 13: OD speed-shift contact normally closed input	0		X
F101	Function selection for control terminal X5	14: CD speed-shift contact normally open input 15: CD speed-shift contact normally closed input 16: MS(multi-section)speed terminal 1	2		X
F102	Function selection for control terminal X6	17: MS(multi-section)speed terminal 2 18: MS(multi-section)speed terminal 3 19: OD prohibiting terminal input	4		X
F103	Function selection for control terminal X7	20: Torque maintaining prohibiting terminal input 21: Low speed OD/CD enabling input 22: Locking signal permanent ON input 23: Locking signal permanent OFF input	1		X
F104	Function selection for programmable relay output A/PB/PC	0: Output signal S1 for completely OD 1: Output signal S1 for completely CD	9	4	X
F105	Function selection for programmable relay output PAC/PC1	2: Output signal S2 for completely OD 3: Output signal S2 for completely CD	0	0	—
F106	Function selection for programmable relay output PAC/PC2	4: Fault relay output 1(POFF state excluded) 5: Fault relay output 2(POFF state included) 6: Output signal S3 for completely OD 7: Output signal S3 for completely CD 8: Locking signal output 9: Door re-opening signal output	1	1	—
F107	Output of relay PC1 and PC2	0: Both PC1 and PC2 do not reverse 1: PC1 reverse 2: PC2 reverse 3: Both PC1 and PC2 reverse	3	0	X
F108	Reserved				

4.10 Display and Monitoring Parameters

F109~F124 User need not modify it.

F110 the value of factory setting is 265.

Chapter 5 Introduction on Detailed Functions

Reference frequency: Target frequency at which the drive operates stably.

Operating frequency: The drive's actual output frequency during operation.

The parameters in “ **【】** ” are this door factory settings of corresponding function codes.

5.1 Basic Operation Function Parameters

F000 User Password

Setting Range: 0~9999 【1234】

Set up any non zero number as user's password to enable the password protection function.

0000: Clear user's password that has been already set up to disable the password protection function.

TD3200 series drive's factory setting is that the password protection function is invalid (F000=0000).

Attention:

Once user's password setting is valid, if you enter parameter setup status again, the parameters can only be read but cannot be revised by keypad, unless you input the password correctly.

F001 Control mode (This door system use Distance control 2)

Setting Range: 0~3 【3】

0: Speed control 1 (this door system do not use)

1: Speed control 2 (this door system do not use)

2: Distance control 1 (this door system do not use)

3: Distance control 2

The drive operates in vector control mode without speed sensor. When the drive operates as a drive controls elevator door, the PG parameters must be set correctly, otherwise the accuracy of door width cannot be ensured. The door width should be obtained via auto-learning, and the door width information should be saved after auto-learning. Besides, the distance control parameters (F049~F052) should be set correctly, during operation, the pulse count will be compared with the preset value of distance control parameters so as to achieve speed changing and complete CD or OD.

In this mode, PG may not be connected to the motor's shaft.

F002 Control commands selection

Setting Range: 0~4 【0】

0: Panel control mode of general-purpose drive (this door system do not use)

1: Terminal control mode of the drive used for controlling elevator door

The drive sends OD and CD commands via control system to realize door opening or closing. The operating logic is shown in Fig. 5-1.

K1	K2	Operating command
0	0	stop
1	0	OD
0	1	CD
1	1	OD

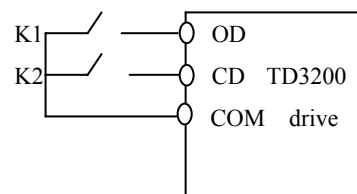


Fig. 5-1 Terminal control logic

2: Manual testing mode

In this mode, the user can test the door operation and set the CD and OD control

signal as well as all the relevant parameters of CD and OD operation curves. Press

RUN and ▲ together to open door, press **RUN** and ▼ together to close door, and press **STOP** to stop the motor.

 Attention:

Door width auto-learning is only valid in manual testing mode. Pressing **RUN** can start the auto-learning.

3: Auto-demo mode

Auto-demo mode is used for demonstrating the operation of the drive and commissioning of the drive inside factory, and the drive is no need to be under the control of the control system. The auto-demo mode can be set after the drive's operation curve is set in manual testing mode. Press **RUN** to start the demonstration of OD and CD repetitively, and the interval between CD and OD can be set via F063 and F064. Pressing **STOP** can stop the door operation.

4: Terminal control mode of general-purpose drive (this door system do not use)

F003 Max. output frequency **Setting Range: 50.00~400.00Hz 【50.00】**

Max. output frequency is the highest frequency of that is able to be output by drive.

F004 Running direction selection **Setting Range: 0、1 【0】**

Motor's running direction can be changed by setting F004. The motor's running direction is determined when the motor is wired correctly. The user can change the setting of F004 to change the direction without the necessity to rewiring the motor.

0: Same with preset direction

1: Reverse to preset direction

F005 Low-speed operation setting **Setting Range: 0.00~50.00Hz 【3.00】**

F005 is used to set the speed of low speed operation that includes the auto-testing operation at start-up, and the operation when the low speed running command is active.

 Note:

Auto-testing operation at start-up: If the drive is set in elevator-door control mode, when it is powered on, it will conduct low-speed CD at first. If it is hindered in CD process, then it will open the door again, and it will close the door again after the door is opened completely. After the door is closed completely, the drive stops and the auto-testing operation is over, and the elevator door control system is in completely-CD status at this time.

F006 Limit of CD or OD time **Setting Range: 0.1~3600s 【300.0】**

F006 is used to set the maximum time of CD and OD operation. The user should set its parameters reasonably according to the actual conditions and **settings must be bigger or equal to the sum of all the settings of CD or OD (parameter group of CD and OD curve)**, otherwise alarm for parameter setting error may occur (E028). Setting F006 correctly can protect the system against abnormal operation of elevator

door. The normal operation time will not exceed the setting of F006, but when the CD or OD position limiting is disabled, which may cause the CD or OD cannot be ended, the operation time may exceed the setting of F006, alarm for CD or OD error may occur (E030).

 Note:

1) OD time: The duration from the time when the drive receives the OD command to the time when the door is opened completely.

2) CD time: The duration from the time when the drive receives the CD command to the time when the door is closed completely.

3) When the drive operates as a general-purpose drive, there is no limit for CD or OD operation time.

F007 Delay time for OD terminal command **Setting Range: 0.0~3600s 【0】**

F007 is used to set the time during which the drive maintains its OD operating status after the terminal

command is disabled. The drive stops when the time is bigger or equal to the setting of F007. If the OD command is invalid during the door is opening, the drive will stop immediately and have no delay function.

F008 Delay time for CD terminal command **Setting Range: 0.0~3600s** **【0】**

F008 is used to set the time during which the drive maintains its CD operating status after the terminal command is disabled. The drive stops when the time is bigger or equal to the setting of F008. If the CD command is invalid during the door is closing, the drive will stop immediately and have no delay function.

5.2 CD and OD Operating Parameters

F009 Display selection for F009~F025 **Setting Range: 0、1** **【0】**

F009 decides whether to display the function parameter group (F010~F025).

0: Not display

1: Display

 Note:

In function codes displaying status, pressing  can scroll through the function codes to be displayed. All the function codes of “Display selection for Function module xxx~xxx” will be displayed first.

F010 OD start torque **Setting Range: 0.0%~150.0%** **【50%】**

F010 is used to set the start torque when the motor starts to open the door so as to ensure the nice OD effects. The value of the torque is the product by multiplying the setting of F010 with the rated torque.

F011 OD startup Acc time **Setting Range: 0.1~3600s** **【1.0s】**

F011 is used to set the time taken for the motor to accelerate from zero speed to low speed in OD startup process.

 Note:

1. Acc time is the time taken for the motor to accelerate from 0Hz (zero speed) to the maximum frequency (maximum speed).
2. Dec time is the time taken for the motor to decelerate from maximum frequency (maximum speed) to 0Hz (zero speed).

F012 Low speed in OD startup process **Setting Range: 0~50.00Hz** **【10.0Hz】**

F012 is used to set the low operating frequency (speed) in OD process.

F013 low-speed maintaining time at OD startup **Setting Range: 0.1~3600s** **【1.0s】**

F013 is used to set the low-speed operating time including the Acc time and constant speed operating time in OD startup process.

F014 OD reference frequency setting **Setting Range: 0~ Maximum frequency** **【35.00】**

F014 is used to set the high operating frequency (speed) in OD process.

F015 OD Acc time **Setting Range: 0.1~3600s** **【2.0s】**

F015 is used to set the time taken for the motor to accelerates from the OD startup low speed to the OD frequency in OD process.

F018 OD Dec time **Setting Range: 0.1~3600s** **【2.0s】**

F018 defines time taken for the motor to decelerate from the OD frequency (high speed) to the low speed in OD ending status in OD process.

F021 low speed setting in OD ending phase **Setting Range: 0.0~50.00Hz** **【3.0】**

F021 defines the motor's low operating frequency (speed) after the OD speed changing signal is active in OD process.

F022 Holding torque in complete OD status **Setting Range: 0.0%~150.0%** **【50.0%】**

When the motor is in OD operating process, the OD location limiting function is enabled, and the torque of catching spinning motor is bigger or equal to the setting of F023, and in the mean time the torque holding prohibition function is disabled, then the motor maintains the elevator door in complete OD status with the

holding torque that is the product by multiplying the setting of F022 and motor's rated torque.

F023 Threshold setting for OD torque changing **Setting Range: 0.0%~150.0% 【50.0%】**

In OD process, if the OD location limiting signal is active or the pulse count for distance control reaches the value obtained by multiplying the setting of F051 by the door width, the motor continues to run at low speed in the ending phase of OD. When the door reaches the limited location, the motor is caught, and the catching motor torque is bigger or equal to the product obtained by multiplying the setting of F023 by rated torque, then the system is in complete OD status and maintaining the current torque until the motor runs reversely or stops.

F024 Abnormal Dec time **Setting Range: 0.0~100s 【0.5s】**

If re-OD command is active, F024 defines the time taken for the motor decelerates from CD operating frequency to zero.

Under the condition of ensuring no over-current in Dec process, F024 should be set as small as possible so as to ensure the fast return of the door.

F025 Reserved

F026 Display selection for F026~F042 **Setting Range: 0、1 【0】**

F026 decides whether F027~F042 can be displayed or not.

0: Not display 1: Display

F027 Start torque for CD **Setting Range: 0.0%~150.0% 【50.0%】**

F027 is used to set the start torque when the motor starts to close the door so as to ensure good CD effects. The value of the torque is the product by multiplying the setting of F027 by the rated torque.

F028 CD startup Acc time **Setting Range: 0.1~3600s 【1.0s】**

F028 is used to set the time taken for the motor to accelerate from zero speed to low speed in CD startup process.

F029 Low speed in CD startup process **Setting Range: 0~50.00Hz 【8.00】**

F029 is used to set the target frequency (speed) of low operation speed in CD process.

F030 Low-speed maintaining time at CD startup **Setting Range: 0.1~3600s 【1.0s】**

F030 is used to set the low-speed operating time including the Acc time and constant speed operating time in CD startup process.

F031 CD reference frequency setting **Setting Range: 0~Maximum frequency 【30.00】**

F031 is used to set the high operating frequency (speed) in CD process.

F032 CD Acc time **Setting Range: 0.1~3600s 【2.0s】**

F032 is used to set the time taken for the motor to accelerates from the CD startup low speed to the CD frequency in CD process.

F035 CD Dec time **Setting Range: 0.1~3600s 【2.0s】**

F035 defines time taken for the motor to decelerate from the CD frequency (high speed) to the low speed in CD ending status in CD process.

F038 Low speed setting in CD ending phase **Setting Range: 0.0~50.00Hz 【2.00Hz】**

F038 defines the motor's low operating frequency (speed) after the CD speed changing signal is active in CD process.

F039 Holding torque in complete CD status **Setting Range: 0.0%~150.0% 【25.0%】**

When the motor is in CD operating process, the CD location limiting function is enabled, and the torque of catching spinning motor is bigger or equal to the setting of F041, and in the mean time the torque holding prohibition function is disabled, then the motor maintains the elevator door in complete CD status with the holding torque that is the product by multiplying the setting of F039 and motor's rated torque.

F040 Hindering torque for CD high speed phase **Setting Range: 0.0%~200.0% 【100.0%】**

In CD high speed process, the motor's output torque is detected, and if the torque is bigger or equal to the

value obtained by multiplying the setting of F040 by the rated torque, the door is re-opened to the complete OD status. During the OD process, the drive will not response to the CD command.

F041 Threshold setting for CD torque changing Setting Range: 0.0%~150.0% 【50.0%】

In CD process, if the CD location limiting signal is active or the pulse count for distance control reaches the value obtained by multiplying the setting of F052 by the door width, the motor continues to run at low speed in the ending phase of CD. When the door reaches the limited location, the motor is caught, and the catching motor torque is bigger or equal to the value obtained by multiplying the setting of F041 by rated torque, then the system is in complete CD status and maintaining the current torque until the motor runs reversely or stops.

F042 Hindering torque for CD low speed end phase Setting Range: 0.0%~200% 【100.0%】

In CD low speed end process, the motor's output torque is detected, and if the torque is bigger or equal to the value obtained by multiplying the setting of F042 by the rated torque, the door is re-opened to the complete OD status. During the OD process, the drive will not response to the CD command.

5.3 Distance Control Parameters

In distance control, based on the door width information got from auto-learning, the CD/OD process, the running pulse is counted in real time and this pulse compared with the preset pulse so as to realize the CD/OD speed changing complete CD and OD handling. The control process is shown in Fig. 5-2.

Distance control includes distance control 1 and distance control 2, and their control logics and running curve are exactly the same. In distance control 1, the PG information is used both in control and pulse counting, and under this condition, the PG must be connected to the motor in the same shaft. In distance control 2, the PG information is only used in pulse counting, and under this condition, the PG needs not to be connected to the motor in the same shaft. (**This door system use Distance control 2**)

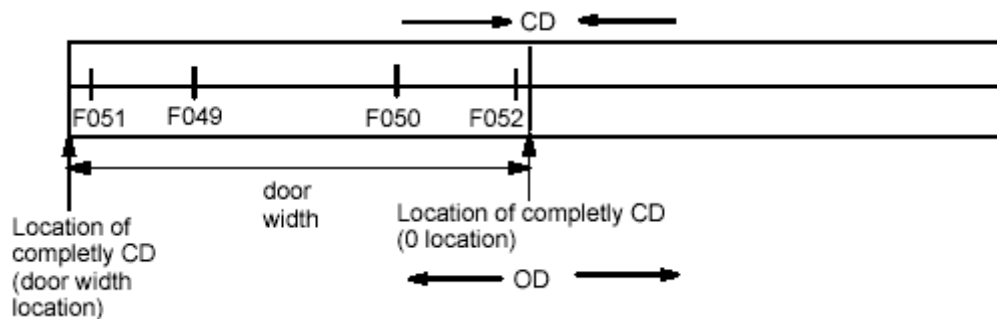


Fig. 5-2 Illustration for distance control

In distance control, besides the operating parameters from F010 to F040 need to be set and adjusted, the parameters from F044 to F052 also need to be set. Door width auto-learning needs to be implemented before distance control. The speed curves of distance control can be illustrated by Fig. 5-3 and Fig. 5-4.

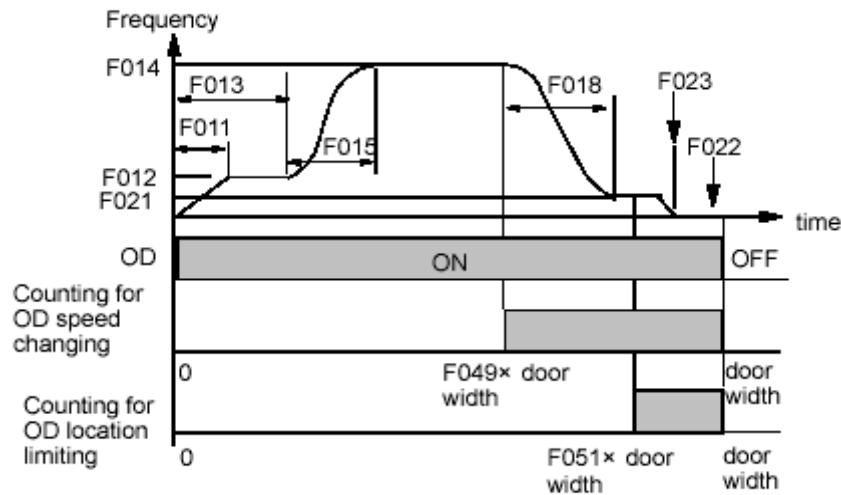


Fig. 5-3 OD operating curve for distance control

OD process for distance control:

- ① When the OD command is active, the motor starts with start torque for OD (F010), and accelerates to low speed in OD startup process (F012) according to OD startup Acc time (F011), and then operates at constant speed in low-speed section.
- ② Start timing when the OD operation starts, when the time is counted to low-speed maintaining time at OD startup (F013), the motor starts to operate at OD reference frequency setting (F014), and accelerates according to OD Acc time (F015), and then operates at high speed after the acceleration.
- ③ When pulse count in OD process $\geq F049 \times \text{door width}$, the motor decelerates according to OD Dec time (F018), and then operates at constant low speed in OD ending phase after the Dec process with low speed setting in OD ending phase (F021) as the target speed.
- ④ When pulse count in OD process $\geq F051 \times \text{door width}$, PAC/PC1 output the signal S1 for complete OD because of F105=0, and the motor continues to run at OD low speed in OD ending phase, and the drive judges the motor's output torque, if the torque is bigger or equal to the threshold setting for OD torque changing (F023), the drive is in OD torque maintaining status, and the torque is equal to the setting of F022, thus the whole OD process is over.
- ⑤ When OD command is disabled, the holding torque in complete OD status is still remain, when the time is bigger or equal to the setting of F007, the drive exits the OD torque maintaining status.

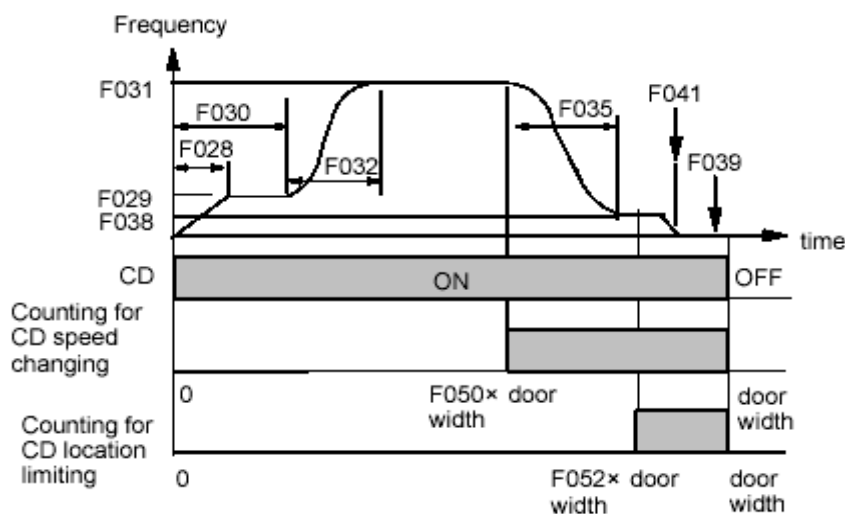


Fig. 5-4 CD operating curve for distance control

CD process for distance control:

- ① When the CD command is active, the motor starts with start torque for CD (F027), and accelerates to low speed in CD startup process (F029) according to CD startup Acc time (F028), and then operates at constant speed in low-speed section.
- ② Start timing when the CD operation starts, when the time is counted to low-speed maintaining time at CD startup (F030), the motor starts to operate at CD high speed (F031), and accelerates according to CD Acc time (F032), and then operates at constant speed in high speed section of CD operation after the acceleration.
- ③ When pulse count in CD process $\geq F050 \times \text{door width}$, the motor decelerates according to CD Dec time (F035), and then operates at constant low speed in CD ending phase after the Dec process with low speed setting in CD ending phase (F038) as the target speed.
- ④ When pulse count in CD process $\geq F052 \times \text{door width}$, PAC/PC2 output the signal S1 for complete CD because of F106=1, and the motor continues to run at CD low speed in CD ending phase, and the drive judges the motor's output torque, if the torque is bigger or equal to the threshold setting for CD torque changing (F041), the drive is in CD torque maintaining status, and the torque is equal to the setting of F039, thus the whole CD process is over.
- ⑤ When CD command is disabled, the holding torque in complete CD status is still remain, when the time is bigger or equal to the setting of F008, the drive exits the CD torque maintaining status.

It is recommended to repower the drive after the auto-learning process is over.

After repowering the drive, the drive tests the running automatically, and stop after the door is closed completely, and the motor is in the initial location (0 bit) of the running counting. At this time, OD/CD operation under distance control can be start.

F043 Display selection for F043~F066 **Setting Range: 0、1 【0】**

F043 decides whether F044~F066 can be displayed or not.

0: Not display

1: Display

F044 Number of pulse per revolution **Setting Range: 1~9999 【100】**

F045 PG phase **Setting Range: 0、1 【0】**

F046 Direction of PG **Setting Range: 0、1 【0】**

TD3200 drive uses the vector control mode, relevant parameters of PG must be set if close loop vector control mode is adopted. In distance control mode, relevant parameters of PG should also be set correctly so as to ensure the good distance control effects.

TD3200 use two-phase PG in this door system.

 Note:

- 1) Two phase PG signals are input via terminal X1 and X2, while signal must be input via X1.
- 2) TD3200 drive only provides 24V power supply, so only PG that power source can be used.
- 3) Use the PG with open collector output or push-pull output.

F047 Speed in door width auto-learning process **Setting Range: 0.0~50.00Hz 【5.0Hz】**

F047 defines the operating frequency in door width auto-learning process. In order to reduce the door width auto-learning error, F047 should not be set to too big so as to avoid the shock when the door is in complete OD or CD status.

F048 Function selection for door-width auto-learning **(Door-width auto-learning is completed in factory, User need not modify)** **Setting Range: 0、1 【0】**

0: Door-width auto-learning is disabled 1: Door-width auto-learning is enabled

Set F048 to 1, press RUN after confirmation, the drive will execute the door-width auto-learning automatically. After the auto-learning, setting of F048 will be resumed to 0 automatically. Process of

door-width auto-learning: After auto-learning is started, the motor operates according to the procedures of CD→OD→CD, after the door is completely closed, the door width information will be saved automatically.

 Note:

- 1) Relevant parameters of PG must be input correctly before the auto-learning.
- 2) Auto-learning is only active in manual testing mode (F002=2).
- 3) During auto-learning process, the rail of the door should not have any object to obstruct the moving of the door.
- 4) If two-phase PG is used, the door width information can be obtained manually without the necessity of operating the motor. Preset F054 as 9999, First, record the pulse number P1 when the door is complete closed, and then pull the door to the completely opened location, record the pulse number P2 at this time. The high and low bits of P1 and P2 can be displayed as stopping parameters, refer to the explanations for F110 and F111.

Door width=|P2 - P1|, and the value is input in F053 and F054 manually.

F049 OD speed changing location under

Setting Range: 60.0%~90.0% (door width) 【70.0%】

In the OD process under distance control, the pulse number is recorded in real time. When the pulse number is bigger or equal to the value obtained by multiplying the setting of F049 by door width, the motor will enter the OD ending phase and operating at low speed.

F050 CD speed changing location under distance control

Setting Range: 60.0%~90.0% (door width) 【70.0%】

In the CD process under distance control, the pulse number is recorded in real time. When the pulse number is bigger or equal to the value obtained by multiplying the setting of F050 by door width, the motor will enter the CD ending phase and operating at low speed.

F051 Pulse setting for complete OD under distance control

Setting Range: 80.0%~99.0% (door width) 【95.0%】

In the OD process under distance control, the pulse number is recorded in real time. When the pulse number is bigger or equal to the value obtained by multiplying the setting of F051 by door width, the drive will handle relevant controls for complete OD, and the same time output the signal S1 for complete OD because of F105=0. The setting of this parameter is a changeless value F51=93%.

F052 Pulse setting for complete CD under distance control

Setting Range: 80.0%~99.0% (door width) 【95.0%】

In the CD process under distance control, the pulse number is recorded in real time. When the pulse number is bigger or equal to the value obtained by multiplying the setting of F052 by door width, the drive will handle relevant controls for complete CD, and the same time output the signal S1 for complete CD because of F106=1. The setting of this parameter is according to Table. 3-4 and Table.3-5

F055~F062 User need not modify it.

5.4 Demo Functions

Demo mode is an auto cycle operating process. Pressing RUN key can start the demo operation, and the motor starts OD operation according to the operation curve. The drive starts timing after the door is opened completely. When the time reaches the setting of F063, the drive starts reverse CD operation automatically, and starts timing after the door is closed completely. When the time reaches the setting of F064, the drive starts reverse OD operation automatically, thus it operates reciprocally in this way, until the STOP key is pressed, and then the drive stops and the demo is ended.

Demo operation can be used for demonstration and burning test.

5.5 Motor's Parameters (These parameters have setted by door factory.)

F067 Display selection for F067~F086 **Setting Range: 0, 1 【0】**

F067 decides whether F068~F086 can be displayed or not.

0: Not display

1: Display

F068 Motor type selection **Setting Range: 0 【0】**

0: Asynchronous motor

F069 Rated power **Setting Range: 0~750W 【150】**

F073 Motor rated speed **Setting Range: 1~9999rpm 【900】**

F072 Motor rated frequency **Setting Range: 1.00~400.00Hz 【50.00】**

F071 Motor rated current **Setting Range: 0.10 ~9.90A 【0.9】**

F070 Motor rated voltage **Setting Range: 0~380V 【220】**

Set the parameters of motor according to the nameplate.

 Note:

- 1) Relevant parameters of motor (F068~F073) must be input correctly before the auto-tuning.
- 2) The motor power should match that of the drive, that is, the setting of F069 can only be one grade higher or two grades lower than standard motor capacity so as to ensure the performance.

F074 Reserved

F075 Auto-tuning for getting motor's parameters **Setting Range: 0, 1 【0】**

0: Auto-tuning function is disabled

1: Auto-tuning function is enabled

Enter the motor parameters according to the nameplate (F069~F073) before tuning. First set F075 to 1, after confirmation, press RUN on the operation panel, the drive will execute auto-tuning function. After tuning completed, setting of F075 will resume to 0 automatically.

 Note:

- 1) Do not start tuning with load on motor;
- 2) Make sure the motor is in stopping status before tuning, otherwise, the tuning can not be processed normally;
- 3) Tuning can be performed in panel control mode only (F002=0, factory default value);
- 4) For the motor of same model and made by same manufacturer, just input the motor parameters according to the nameplate, and auto-tuning is not needed.

F076 Stator resistance **Setting Range: 0.00~99.99 Ω 【7.73】**

F077 Stator inductance **Setting Range: 0~9999mH 【357】**

F078 Rotor resistance **Setting Range: 0.00~99.99 Ω 【5.23】**

F079 Rotor inductance **Setting Range: 0~9999mH 【357】**

F080 Mutual inductance **Setting Range: 0~9999mH 【325】**

F081 Magnetising current with no load **Setting Range: 0.00~99.99A 【1.08】**

After auto-tuning is over, settings of F076~F081 will be updated. Each time after changing the values on the motor's nameplate, the settings of F076~F081 will be set as default values by the drive. Meanings of parameters are shown in Fig. 5-5.

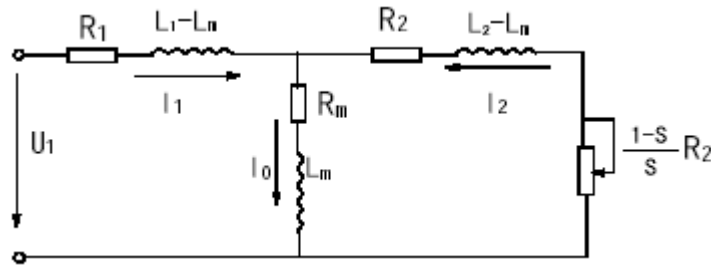


Fig. 5-5 Steady-state equivalent circuit of asynchronous motor

In Fig. 5-5, R_1 , L_1 , R_2 , L_2 , L_m , I_0 represent stator resistance, stator inductance, rotor resistance, rotor inductance, mutual inductance and magnetising current respectively.

5.6 Auxiliary Parameters

F082 ~F086 User need not modify it.

5.7 Vector Control Parameters

F087 ~F095 User need not modify it.

5.8 Digital I/O

F096 Display selection for F096~F108

Setting Range: 0、1 【1】

F096 decides whether F097~F108 can be displayed or not.

0: Not display

1: Display

Function and parameters of control terminal X1~X7 are defined by the following function codes.

F097 Function selection for control terminal X1 Setting Range: 0~23 【10】

F098 Function selection for control terminal X2 Setting Range: 0~23 【11】

F099 Function selection for control terminal X3 Setting Range: 0~23 【0】

F100 Function selection for control terminal X4 Setting Range: 0~23 【0】

F101 Function selection for control terminal X5 Setting Range: 0~23 【2】

F102 Function selection for control terminal X6 Setting Range: 0~23 【4】

F103 Function selection for control terminal X7 Setting Range: 0~23 【1】

Control terminals X1~X7 are programmable input terminals, and their functions can be defined by setting the values of F097~F103.

0: No function

1: External reset (RESET) signal input When the drive alarms for fault, the fault can be reset via this terminal. This enabled at the rising edge of pulse, and its function is consistent with that of **RESET** key on the panel.

2/3: Normally open/closed input contacts for optic signal In CD process, if this terminal is activated, then the OD command is re-executed for protection, after the door is re-opened completely, then next CD command can be executed. During the re-opening process, the drive will not response to CD command. If the door has reached the CD location limit, this protection signal is ineffective.

4/5: Normally open/closed input contacts for touching board In CD process, if this terminal is activated, then the OD command is re-executed for protection, after the door is re-opened completely, then next CD command can be executed. During the re-opening process, the drive will not response to CD command. If the door has reached the CD location limit, this protection signal is ineffective.

6/7: Normally open/closed input for OD location limiting In OD process under speed control, the drive executes relevant commands about OD location limiting after this signal is enabled.

8/9: Normally open/closed input for CD location limiting In CD process under speed control, the drive execute relevant commands about CD location limiting after this signal is enabled.

10/11: Phase A/B PG signal input (only X1 and X2) Enable the terminal to have the function of inputting PG signal so as to achieve close loop vector control or distance control. Only F097 and F098 can be set to the functions of inputting PG signal.

12/13: Normally open/closed input for OD speed changing In OD process under speed control, the motor starts low speed operation in ending phase after the normally open/closed input for OD speed changing is enabled.

14/15: Normally open/closed input for CD speed changing In CD process under speed control, the motor starts low speed operation in ending phase after the normally open/closed input for CD speed changing is enabled.

16~18: Terminal for inputting multi-section speed operation signal Through the ON/OFF combination of the terminals, up to 8 levels of speed can be defined, and these speed levels can be set by F055~F062 respectively.

19: Terminal for inputting OD prohibition signal OD command will not be affected if this signal is enabled.

20: Terminal for inputting torque-maintaining prohibition signal In CD/OD torque-maintaining process, the drive will maintain the 0 torque if this signal is enabled.

21: Low speed OD/CD enabling signal input If this signal is enabled, the motor operates at low speed according to the speed set by F005.

22/23: Normally open/closed input for locking signal

F104 Function selection for programmable relay output PA/PB/PC Setting Range: 0~9 [9]

F105 Function selection for programmable relay output PAC/PC1 Setting Range: 0~9 [0]

F106 Function selection for programmable relay output PAC/PC2 Setting Range: 0~9 [1]

Notes about the functions:

0: Output signal S1 for complete OD

In OD process, when the drive pulse count number reaches the OD location limiting value (the setting of F51), the drive will output signal S1 for complete OD.

Complete OD signal is disabled in demo mode, general-purpose drive operation mode and CD process.

1: Output signal S1 for complete CD

In CD process, when the drive pulse count number reaches the CD location limiting value (the setting of F52), the drive will output signal S1 for complete CD.

Complete CD signal is disabled in demo mode, general-purpose drive operation mode and OD process.

2: Output signal S2 for complete OD

In OD process, when the drive pulse count number reaches the OD location limiting value (the setting of F51), and at the same time the catching motor torque reaches the setting of F023, the drive will output signal S2 for complete OD.

Complete OD signal is disabled in demo mode, general-purpose drive operation mode and CD process.

3: Output signal S2 for complete CD

In CD process, when the drive pulse count number reaches the CD location limiting value (the setting of F52), and at the same time the catching motor torque reaches the setting of F041, the drive will output signal S2 for complete CD.

Complete CD signal is disabled in demo mode, general-purpose drive operation mode and OD process.

4: Fault relay output G1

When the drive has fault, it will output fault relay output signal G1. But when parameter setting error (E028)

occurs, the drive only displays the alarm information and will not output fault relay output signal 1.

5: Fault relay output G2

When the drive has fault or the drive's DC bus voltage is low (LED displays "P.oFF"), it will output fault relay output signal G2. But when parameter setting error (E028) occurs, the drive only displays the alarm information and will not output fault relay output signal 2.

6: Output signal S3 for complete OD

In OD process, when the drive receives OD location limiting signal or when the pulse count number reaches the OD location limiting value, locking signal is invalid and at the same time the catching motor torque reaches the setting of F023, the drive will output signal S3 for complete OD.

Complete OD signal is disabled in demo mode, general-purpose drive operation mode and CD process.

7: Output signal S3 for complete CD

In CD process, when the drive pulse count number reaches the CD location limiting value (the setting of F51), locking signal is disabled and at the same time the catching motor torque reaches the setting of F041, the drive will output signal S3 for complete CD.

Complete CD signal is disabled in demo mode, general-purpose drive operation mode and OD process.

8: Locking signal output

Synchronously input/output locking signal.

9: Door re-opening signal output

In door re-opening process, output the door re-opening signal.

F107 Output of relay PC1 and PC2

Setting Range: 0~3 【3】

0: Neither PC1 nor PC2 reverse, the output of relay PC1 and PC2 are normally open

3: Both PC1 and PC2 reverse, the output of relay PC1 and PC2 are normally open (**This door system use choice 3, Show in Fig. 3-1**)

5.9 Display and Monitoring Functions

F109~F124 User need not modify it.

Chapter 6 Troubleshooting

TD3200 has 16 kinds of faults. When fault occurs, the user can get the information about the fault codes and the DC bus voltage, output current, operating frequency and terminal status by looking up the values of F112~F119. The user can check the faults according to the following table himself and record detailed fault phenomena before seeking service. Please contact the sales person when seeking service.

Fault code	Type of faults	Possible fault reasons	Troubleshooting
E001	Over-current in Acc process	1) Acc time is too short. 2) Restart the motor in motion when momentary stop occurs. 3) Incorrect external wiring 4) Incorrect wiring of PG	1) Prolong the Acc time; 2) Start the motor after it stops. 3) Wiring correctly 4) Change the setting of F046 or the wiring of PG
E002	Dec overcurrent	Dec time is too short.	Please prolong Dec time Adjust PI parameters
E003	Overcurrent at constant speed operation	1) Sudden change of load 2) Abnormal load	1) Adjust PI parameters 2) Check load
E004	Overvoltage in Acc	1) Abnormal input voltage	1) Please check input power source.

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	process	2) Unsuitable setting of PI parameters 3) Big load inertia	2) Please Adjust PI parameters 3) Connect external braking resistor
E005	Overvoltage in Dec process	1) Dec time is short 2) Unsuitable selection of braking resistor 3) Abnormal input voltage 4) Unsuitable setting of PI parameters	1) Prolong Dec time 2) Re-select the braking resistor 3) Check input voltage 4) Adjust PI parameters
E006	Overvoltage at constant speed operation	1) Abnormal change of input voltage 2) Energy regenerated by negative torque load	1) Mount input reactor 2) Connect external braking resistor
E007 E008	Reserved		
E009	Output phase-failure	The output cables of the drive are broken, phase failure or 3-phase loads are severely unbalanced.	The output cables of the drive are broken, phase failure or 3-phase loads are severely unbalanced.
E010	Reserved		
E011	Overheat of heatsink	(1) Air duct blocked (2) IGBT abnormal	(1) Clear the air duct (2) Seek service
E012	Reserved		
E013	Drive overload	1) Heavy load 2) Very low mains voltage 3) PG error	1) Select drives with bigger ratings 2) Check mains voltage 3) Check the PG wiring or replace the PG
E014 E015	Reserved		
E016	EEPROM read or write fault	Fault occurs in the read-write of control parameters	Seek service
E017 E018	Reserved		
E019	Current detecting circuit fault	(1) Current detecting circuit fails (2) Power source fails	Seek service
E020	CPU fault	(1) Severe interference (2) Fault in read-write of CPU of control board	Seek service
E021 E022 E023	Reserved		
E024	Motor-parameters tuning error	(1) Motor's parameters setting error (2) Tuning overtime	(1) Set the motor parameters correctly according to the motor nameplate (2) Check if the motor is broken away from its load and check the motor's cables
E025 E026 E027	Reserved		

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E028	Parameters setting error	(1)OD width auto-learning is not set as manual testing mode (2)General panel operating mode is not used in motor parameter tuning process (3)CD and OD time is set shorter than all the settings of CD and OD time	(1)Set F002 to 2 in OD width auto-learning (2)Set F002 to 0 in motor parameters tuning (3)Set the OD and CD time to a bigger value and ensure the setting of F006 is bigger than all the settings of CD and OD time.
E029	OD width auto-learning error	(1)The OD width obtained from the auto-learning process is 0. (2)The OD width obtained from the auto-learning process is out of the limits.	(1)Check the PG wiring and relevant parameters (2)Check the mechanical system of the elevator door
E030	OD/CD error	(1)The motor' s running direction is reverse with the definition of CD and OD (2)OD/CD contacts error or setting error (3)PG cable is broken (4)OD width auto-learning overtime	(1)Change the phase rotation of the motor or set F004 to (2)Check OD/CD contacts signal (3)Check PG cable (4)Check the mechanical system of elevator door